Vision Retargeting Using Robot Aware Losses

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Abstract

Monocular vision-based methods for kinematic retargeting from human to robot hands have lagged behind motion capture and glove-based methods. In this paper, we present an improved vision-based retargeting method that incorporates retargeting loss functions into the training of a 3D hand pose estimation model. Instead of learning only keypoint locations, our model also learns to optimize for a desirable retargeted robotic hand pose. As a result, our method improves upon the downstream retargeting results compared to a baseline 3D pose estimation model, beating the baseline pinch success rate on the Allegro Hand by 5.9%. Our code is available at https://github.com/JM5064/vision-retargeting.

1 Introduction

Kinematic retargeting is a persistent challenge in robotics research and a central problem in dexterous manipulation. Such systems are essential for human teleoperation of robots and require accurate and intuitive mappings from human to robot hands. However, finding a natural mapping is inherently difficult due to differences in kinematic structure, joint limits, and degrees of freedom between the two systems. Previous works have devised solutions to kinematic retargeting by mapping keypoints on the human hand to robot joint configurations. To obtain the shape of the human hand, most existing work utilizes deep neural networks for keypoint prediction [4, 14], motion capture, or glove-based solutions [17, 18]. In this paper, we attempt to tackle the quality of vision-based retargeting techniques. Due to the ill-defined problem of inferring 3D keypoints from a 2D image, obtaining high-quality 3D hands is challenging. As a result, vision-based solutions suffer from noisy keypoints, leading to less reliable retargeting. Nevertheless, vision-based solutions are desirable as they provide a low-cost alternative to motion capture and glove-based solutions.

In this work, we investigate improving vision-based retargeting. While standard 3D hand pose estimation models can effectively reproduce the structure of a human hand, they are typically robot-agnostic. As a result, predictions are susceptible to noise, and small errors can lead to poor retargeting quality. To solve this, we propose a robot aware alternative which introduces retargeting objectives directly into the training of a 3D hand pose estimator. Specifically, we incorporate retargeting loss functions that encourage our model to predict keypoints that produce accurate robotic hand shapes. By optimizing for both pose and retargeting accuracy during training, our model is better suited for kinematic retargeting than a standard 3D pose estimation model.

2 Related Work

2.1 3D Hand Pose Estimation

Large strides have been made in pose estimation over the last decade. Several methods used for 2D pose estimation [7, 16] have been adapted to the 3D case [3, 6, 19]. The problem of inferring 3D keypoints from a 2D image is not well defined. However, as humans have physical constraints on how our bones are positioned and can move, deep learning models are able to utilize large datasets [20] to

accurately infer 3D. Methods for pose estimation typically fall into three categories: regression-based approaches [1, 15, 19], heatmap-based approaches [7, 8, 16], and model-based approaches [11, 12].

Regression-based approaches utilize powerful feature extractors to directly predict the locations of keypoints. With the rise of the transformer architecture, regression-based methods have become the dominant method for pose estimation.

Heatmap-based approaches produce keypoint heatmaps instead of directly predicting joint locations. Typically, heatmaps are generated per-joint as a Gaussian bubble over the true location of the keypoint. This method works well with convolution-based methods, as convolutions excel at learning spatial relationships between keypoints.

Model-based approaches incorporate parametric hand models [12] to predict hand parameters instead of keypoints. This has the advantage of constraining predictions to physically plausible hand shapes.

For our work, we utilize a heatmap-based approach to predict 3D keypoints, as they are easily interpretable and have been shown to work reliably.

2.2 Kinematic Retargeting

Kinematic retargeting from human to robot hands is the problem of mapping human hand poses to robot joint configurations. Previous works have tackled this challenge by forcing the joint locations and directions of the retargeted hand to be similar to those of the human hand [4, 17]. Other works have proposed objectives that optimize motion similarity [18].

Vision-based retargeting methods aim to remove the need for specialized hardware by predicting human hand pose directly from images. For example, DexPilot [4] employs a multi-camera setup to track and predict 3D hand keypoints. Later, [14] simplified their setup by utilizing large-scale, unlabeled internet data to perform retargeting through a monocular camera. However, these approaches are sensitive to noise in 3D pose estimation, which can degrade retargeting performance.

Our work differs from prior approaches by integrating retargeting objectives directly into the training of a 3D pose estimation model. Instead of applying retargeting merely as a post-processing step, we incorporate retargeting losses into the learning process, encouraging predicted keypoints to produce consistent robot hand configurations.

3 Method

3.1 Marginal Heatmaps

Heatmaps along with convolutional neural networks have been proven to work well with 2D pose estimation [7, 16] as they take advantage of a CNN's ability to localize joints. To extend this idea into 3D, many papers employ 3D volumetric heatmaps [10] or use depth maps [5] to predict the z value. Full 3D volumetric heatmaps, however, are memory intensive and lead to high computational costs.

The idea of 2D marginal heatmaps was introduced by Nibali et al. [8]. Instead of predicting 3D volumetric heatmaps, we predict 2D xy, xz, and zy heatmaps for each joint. This cuts down the memory cost drastically, making it analogous to 2D pose estimation. Each heatmap is generated by placing a Gaussian bubble with $\sigma = 1$ at the ground truth joint position.

We calculate 3D keypoints from the heatmaps using soft-argmax. This method converts each heatmap into a probability distribution by applying softmax, and then calculates the expected value of each heatmap to predict keypoint location. As a result, we are able to obtain fine-grain keypoint predictions through a differentiable method. Like [8], we use the xy heatmap to predict the 2D location of each joint, and then calculate the z location by averaging the expected values of the xz and zy heatmaps.

In addition, we impose a Jensen-Shannon divergence term also used in [8] to train our heatmaps. This forces the heatmaps to maintain a Gaussian shaped distribution over each keypoint and produces interpretable results.

We calculate the pose loss $\mathcal{L}_{\text{pose}}$ as a sum of the Mean Squared Error between the predicted and ground truth keypoints, and the Jensen-Shannon divergence between the predicted and ground truth heatmaps.

3.2 Retargeting Losses

In addition to keypoint and heatmap losses used for 3D pose estimation, we introduce retargeting losses designed to predict robot-compatible hand shapes. We build upon the objectives proposed by Xin et al. [17] and adapt them for our training.

Let x_i and $\hat{x}_i$ denote the ground truth position and predicted position for the ith fingertip, respectively. We map human keypoints to robot joint configurations using a differentiable inverse kinematics function f, and obtain robot keypoints via forward kinematics:

$$p_i = \text{FK} \circ f(x_i), \quad \hat{p}_i = \text{FK} \circ f(\hat{x}_i)$$

where p_i and $\hat{p}_i$ represent the robot position of the retargeted ground truth and predicted keypoints of the ith fingertip respectively.

Unlike [17], we compare the shape of the retargeted prediction with the retargeted ground truth. This allows us to directly compare robot positions in task-space. Thus, we define the following retargeting losses:

Hand Pose Loss:

$$\mathcal{L}_{\text{hand_pose}} = \|p_{\text{thumb}} - \hat{p}_{\text{thumb}}\|_2^2$$

Hand Shape Loss:

$$\mathcal{L}_{\text{hand_shape}} = \sum_{i=1}^{N} \tilde{s}(d_i)\|v_i - \hat{v}_i\|_2^2$$

Pinch Loss:

$$\mathcal{L}_{\text{pinch}} = \sum_{i=1}^{N-1} s(d_i)\|\gamma_i - \hat{\gamma}_i\|_2^2$$

Fingertip Orientation Loss:

$$\mathcal{L}_{\text{orientation}} = \sum_{i=1}^{N} \|r_i - \hat{r}_i\|_2^2$$

For each fingertip i, v_i represents the task-space vector between the ith fingertip and wrist, γ_i the task-space vector between the ith fingertip and thumb, and r_i the task-space vector between the ith fingertip and distal interphalangeal (DIP) joint:

$$\begin{aligned} v_i &= p_i - p_{\text{wrist}}, \quad \hat{v}_i = \hat{p}_i - \hat{p}_{\text{wrist}} \\ \gamma_i &= p_i - p_{\text{thumb}}, \quad \hat{\gamma}_i = \hat{p}_i - \hat{p}_{\text{thumb}} \\ r_i &= p_i - p_{\text{DIP},i}, \quad \hat{r}_i = \hat{p}_i - \hat{p}_{\text{DIP},i} \end{aligned}$$

Here, $d_i = \|\gamma_i\|$ represents the distance between fingertip i and the thumb. We utilize switching weight functions $s(d_i)$ and $\tilde{s}(d_i)$, which are defined as $\text{sigmoid}(d_i, \epsilon, 10)$ and $\text{sigmoid}(d_i, \epsilon, -10)$ respectively. This allows the pinch loss to be high when finger i is close to the thumb, and the hand shape loss to be high otherwise.

We calculate our retargeting loss as a weighted sum of the above terms:

$$\mathcal{L}_{\text{retargeting}} = \lambda_1 \mathcal{L}_{\text{hand_pose}} + \lambda_2 \mathcal{L}_{\text{hand_shape}} + \lambda_3 \mathcal{L}_{\text{pinch}} + \lambda_4 \mathcal{L}_{\text{orientation}}$$

For our implementation, we utilize a weight scheduling method where each λ_i increases its weight during training.

Our differentiable inverse kinematics function f is implemented following GeoRT [18], allowing gradients to flow from task-space back to the heatmaps.

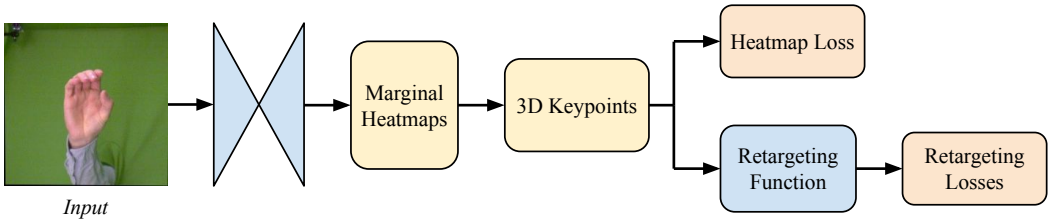

Figure 1: An overview of the training pipeline. An image is fed into a CNN model, which produces marginal heatmap predictions for each joint. We obtain keypoints from these heatmaps using soft-argmax and calculate heatmap losses. Then, we pass our keypoints into our retargeting function to calculate retargeting losses.

3.3 Pipeline Design

Figure 1 illustrates our training procedure:

1. An input image is fed into our CNN model, where it is processed and produces a set of marginal heatmaps xy, xz, zy for each keypoint.
2. We extract predicted 3D keypoints from the heatmaps using soft-argmax
3. We compare the predicted keypoints and heatmaps against the ground to calculate heatmap losses.
4. We apply differentiable inverse kinematics f and forward kinematics to map the predicted keypoints and ground truth keypoints into robot joint positions.
5. We compare the results to calculate retargeting losses.

We formulate our total loss function as a weighted sum of pose losses and retargeting losses:

$$\alpha \mathcal{L}_{\text{pose}} + \beta \mathcal{L}_{\text{retargeting}}$$

For heatmap prediction, we use a ResNet backbone pretrained on ImageNet [13] and MPII [2] to perform feature extraction. We attach three upsampling heads similar to [16] to predict the marginal heatmaps.

4 Experimental Setup

To validate our method, we train two pose estimation models:

- A baseline model trained with marginal heatmaps and soft-argmax supervision.
- A retargeting model trained with the same, along with the retargeting losses.

We train both models on the FreiHAND dataset [20]. For the training set, we utilize FreiHAND's green screen images as well as the augmented background images. Due to compute constraints, we partition our training set into 8 equally sized chunks. During training, we iterate over these chunks and perform validation after each chunk. Once all 8 chunks are completed, we reshuffle the dataset and train on another 8 chunks. We continue this process for a total of 75 chunks. For data augmentation, we utilize random rotation of $\pm$ 45 degrees and color jitter. Testing is performed on the FreiHAND evaluation set.

For training the differentiable IK model, we used GeoRT's retargeting algorithm [18] and trained it for 30 epochs on the FreiHAND dataset.

To validate our model on pose estimation, we calculate PCK@20mm and Mean Per Joint Position Error. PCK@20mm reports the percentage of joints that are predicted within 20 millimeters of the true location. Mean Per Joint Position Error (MPJPE) calculates the average Euclidean error for each joint and is reported in millimeters.

Table 1: 3D pose estimation results on the FreiHAND test dataset.

Model	PCK@20mm ↑	MPJPE ↓
Baseline	**75.3**	**17.0**
Retargeting	74.8	17.1

Table 2: Retargeting results on the FreiHAND test dataset. Pinch success rate is calculated with a 50mm threshold between robot fingertips.

Model	Average Pinch Distance (mm) ↓	Pinch Success Rate (%) ↑
Baseline	57.8	49.4
Retargeting	**54.7**	**55.3**

To validate our model on kinematic retargeting, we calculate the robot's average pinch distance and pinch success rate. Pinching is essential to determining a good retargeting, as many difficult tasks require close pinching of the fingers. We thus evaluate our method by determining how well it performs pinching: if a human hand is pinching, we want the retargeted robot hand to also be pinching.

For each finger, we say that it is pinching if the distance from the fingertip to the thumb is less than a given threshold. For our experiments, we use 30 millimeters as the threshold for human hand pinching and 50 millimeters as the threshold for Allegro Hand pinching.

Average pinch distance is defined as the distance between the robot's thumb and pinching fingers, averaged over all samples where the human hand is pinching. Pinch success rate counts the percentage of actual pinches that the robot performs correctly.

5 Results

5.1 Quantitative Results

We first compare the effect of the retargeting losses on pose accuracy. Table 1 shows the hand pose estimation performance between our baseline model and our retargeting model. We see that despite the additional loss functions, the retargeting model is still able to perform competitively on pose estimation. However, we observe that there is a tradeoff to introducing retargeting loss functions, as performance on 3D pose estimation is slightly worse compared to our base model. This result is expected, as our model is prioritizing achieving a more accurate robot configuration over pose metrics.

We then compare the effect of retargeting losses on kinematic retargeting. Table 2 shows the effect of retargeting losses on pinching. We observe that the retargeting model performs better than the baseline. We achieves a lower average pinch distance by 3.1mm, and improve pinch success rate by 5.9%.

5.2 Qualitative Results

In addition to quantitative results, we evaluate our model qualitatively to analyze retargeting results. Figure 2 shows the examples of retargeted results for both models. We observe that overall, the model trained with retargeting losses performs pinching with a higher degree of success, while maintaining overall hand shape for both pinching and non-pinching hands.

Table 3: Pose estimation and retargeting results on the FreiHAND test dataset between our retargeting model and the retargeting model with distance rescaling function $l(d_i)$.

Model	MPJPE ↓	Average Pinch Distance (mm) ↓	Pinch Success Rate (%) ↑
Retargeting	**17.1**	**54.7**	**55.3**
Retargeting + $l(d_i)$	17.2	55.6	53.6

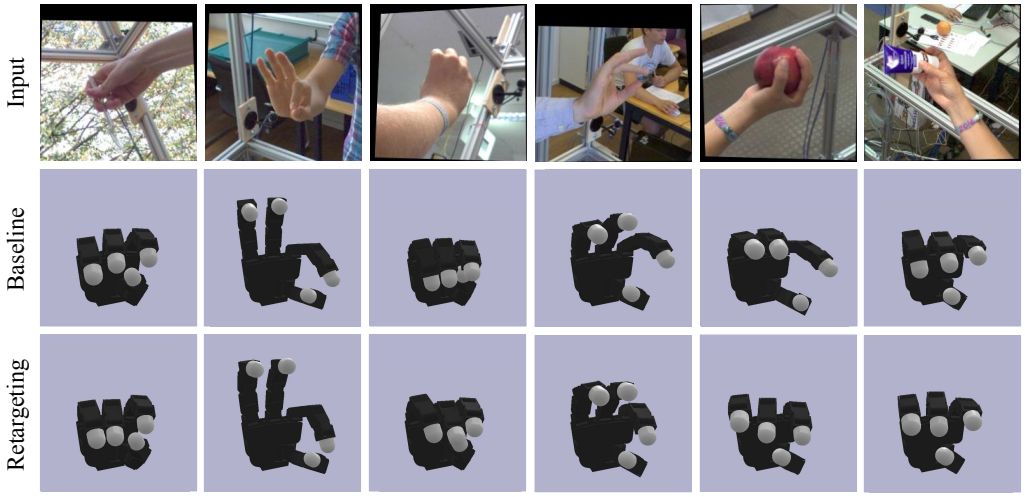

Figure 2: Qualitative results of the baseline model vs. retargeting model on the Allegro Hand. The first four images display pinching actions, and the last two display hands holding objects. We observe that the retargeting model performs better than the baseline model in terms of both hand shape and pinch.

However, we also observe that retargeting results are much more dependent on the retargeting function used to process the keypoints. When using a retargeting model trained on a different distribution of data or with different normalization, the retargeting results differ greatly. Therefore, by utilizing a strong retargeting function, our method serves as an upgrade for vision-based retargeting systems.

5.3 Rescaling Function Analysis

We analyze the effect of adding a distance rescaling function $l(d_i)$ to our pinch loss. This loss is used in [4, 9] to create a smooth transition between pinching and non pinching hands. Our pinch loss with the distance rescaling function is as follows:

$$\mathcal{L}_{\text{pinch_rescaled}} = \sum_{i=1}^{N-1} s(d_i) \| \gamma_i - \frac{l(d_i)}{d_i} \hat{\gamma}_i \|_2^2$$

where $l(d_i)$ is defined as:

$$l(d_i) = \begin{cases} 0, & d_i < \epsilon_2 \\ \frac{\epsilon_1}{\epsilon_1 - \epsilon_2}(d_i - \epsilon_2), & \epsilon_2 \leq d_i \leq \epsilon_1 \\ d_i, & d_i > \epsilon_1 \end{cases}$$

This maps the interval $[\epsilon_2, \epsilon_1]$, a region where the robot hand is considered to be pinching, into $[0, \epsilon_1]$. This encourages a stronger pinch when the thumb-finger distance is below ϵ_2.

Table 3 shows the pose estimation and retargeting results for the models trained with and without the distance rescaling term $l(d_i)$. We find that including the rescaling term hinders the model in both pose estimation and pinching.

We hypothesize that this is due to our direct task-space comparison approach. Unlike other works that utilize this rescaling term to map a human hand pinch to a precise robot pinch, our method directly compares a retargeted ground truth hand to a retargeted prediction. Therefore, this rescaling term becomes redundant and introduces noise into the learning process.

6 Conclusion

We present a vision-based retargeting method by introducing retargeting objectives directly into the training of a 3D hand pose estimation model. By integrating differentiable retargeting losses into the training process, our approach produces predicted hand poses that are more suited for dexterous manipulation tasks. Overall, our work demonstrates that incorporating retargeting objectives during training is a simple yet effective way to improve vision-based methods for kinematic retargeting. For future work, we turn to model-based methods for 3D hand pose estimation. Model-based methods predict hand model parameters instead of keypoint locations. This method aligns well with kinematic retargeting, as both tackle the challenge of achieving natural hand shapes. Thus, combined with our robot aware losses, a model-based pose estimation method could produce favorable retargeting results while maintaining the shape of a hand.

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